

FraudLens — Project Structure

```
FraudLens/
├─ app.py                # Main Streamlit entry point – Try It page,
loads model, handles predictions
├─ pages/
|   └─ 1_Model_Performance.py # Second Streamlit page – metrics, confusion
matrix, methodology
├─ requirements.txt      # Pinned dependencies for local + Streamlit
Cloud
├─ README.md             # Project overview, live demo link,
screenshots, how to run
├─ .gitignore            # Excludes data/raw/, venv/,
.ipynb_checkpoints/, __pycache__/
├─ .streamlit/
|   └─ config.toml        # Theming (colors, layout) for the deployed
app
|
├─ data/
|   └─ raw/                # creditcard.csv lives here – gitignored (too
large, download instructions in README)
|
├─ notebooks/
|   └─ 01_eda.ipynb        # Day 3 – exploratory data analysis
|   └─ 02_preprocessing.ipynb # Day 4 – scaling, splitting, imbalance
strategies
|   └─ 03_model_training.ipynb # Day 5-6 – training, evaluation, model
selection
|
├─ src/
|   └─ __init__.py
|   └─ data_utils.py      # load_data, scale_features, split_data,
apply_smote
|   └─ model_utils.py     # train_model, evaluate_model
|   └─ explain_utils.py   # get_top_factors
|
├─ models/
|   └─ fraud_model.pkl     # Final trained model (Day 6) – committed,
small enough for git
|   └─ scaler.pkl          # Matching fitted scaler – committed
alongside the model
|   └─ MODEL_CARD.md       # Algorithm, strategy, metrics, training
```

```

date, limitations
|
├─ docs/
|   ├── PRD.md                # Or FraudLens_PRD.docx
|   ├── IMPLEMENTATION_BLUEPRINT.md
|   ├── ARCHITECTURE.md
|   ├── SCHEMA.md
|   ├── API.md
|   ├── UI-WIREFRAMES.md
|   ├── PROJECT-STRUCTURE.md  # this file
|   └─ pitch-deck.pdf
|
└─ tests/
    └─ test_model_utils.py    # Basic sanity tests (Day 9)

```

Why this structure

- **notebooks/** vs **src/**: Notebooks are for exploration and are allowed to be messy; anything the app actually depends on gets extracted into **src/** as clean, importable, testable functions. This separation is a standard data science practice and prevents **app.py** from importing fragile notebook code.
- **models/** is **version-controlled (not gitignored)**: Unlike the raw dataset (~150MB, excluded), the trained model artifacts are small (a few MB at most) and *must* be in the repo — Streamlit Community Cloud has no separate model storage step; it deploys straight from the git repo.
- **docs/** **centralizes every planning artifact** produced across the 10 days, so the repo tells the full story of the build process, not just the final code — this is deliberate for recruiter visibility into process, not just output.
- **pages/** **follows Streamlit's built-in multipage convention** (**pages/** folder with numbered files) rather than a custom routing solution — zero extra dependencies, works out of the box on Streamlit Cloud.
- **tests/** is **minimal by design** — full test coverage is out of scope per the PRD's time budget; a few sanity tests on **model_utils.py** are enough to catch the most damaging class of bug (a broken prediction pipeline) before deployment.

What's NOT here (and why)

- No **backend/**, **frontend/**, or **api/** folders — there's no client-server split (see ARCHITECTURE.md).
- No **migrations/** or **database/** folder — no database in v1.0.
- No **auth/** folder — no authentication in v1.0.